

JORDI FABREGAT

Gameplay Programmer · Unreal Engine 5 / Unity · C++ / C#

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[Portfolio](#) · [itch.io](#) · [GitHub](#) · [LinkedIn](#)

PROFILE

Gameplay programmer who ships finished games. Six months solo-building Do Not Follow, a first-person horror game in Unreal Engine 5 (C++), has given me hands-on production experience with UE5 gameplay programming, AI, and rendering — on top of three completed Unity/C++ projects and professional engineering at a research lab. I work the full stack, from high-level gameplay to low-level engine subsystems, and ship reliable, save-aware systems. Final-year Computer Engineering student at UPF, ready to contribute production code from day one.

SELECTED PROJECTS

Do Not Follow — Solo · Unreal Engine 5 / C++ · 2025 – present (~6 months)

A first-person horror game built solo in Unreal Engine 5 — ongoing, with a full vertical slice of mechanics complete.

- Architected the entire C++ gameplay framework around reusable interfaces (IInteractable, ISaveable): a full save/load system, a 9-slot inventory + hotbar, flashlight/battery, stamina, and held-item swapping, all built as save-aware actors.
- Built the enemy AI in C++ — a three-state controller (Patrol / Chase / Investigate) using UE5 AI Perception for both sight and hearing, with player noise events scaled by crouch/walk/sprint over a baked navmesh.
- Engineered a dozen-plus custom C++ gameplay actors (drivable vehicles, sequence puzzles, a pulley crate, AI-summoning trigger zones, a mash-duel door) and a photo mechanic using SceneCaptureComponent2D render targets with line-traced reveals.
- Authored a global post-process pixelation material (parameter-collection driven for runtime tuning) and scripted Sequencer cutscenes and level streaming for the opening and ending.

Alphinland — Solo · Unity / C# · 2025

[trailer](#) [play](#)

A first-person narrative walking sim about grief, built solo end-to-end in one month.

- Architected and shipped every gameplay system in C# — interaction, progression, and scene flow — taking a blank Unity project to a polished vertical slice.
- Built a three-tier interaction system with an inspect mode that spawns an isolated, free-rotating instance of any object, plus a dual-world transition driven by a custom post-processing shader.

Co-op Mixed-Reality Tower Defense — Team · Unity / C# · HTC Vive · 2024

- Coded a forced-cooperation resource mechanic requiring synchronised input from both players, plus wave-spawning and tower-placement systems with difficulty scaling.
- Owned the balancing pass across repeated playtests. Work led to selection into UPF's Full-Body Interaction Lab.

EXPERIENCE

Gameplay Programmer (R&D Assistant) · Full-Body Interaction Lab, UPF

Sep 2025 – Feb 2026

- Rebuilt and extended the Unity codebases for DragonIce and Lands of Fog — full-body mixed-reality games (6×6 m projected playspace, HTC Vive trackers). Lands of Fog is a published, RecerCaixa-funded research system.
- Eliminated critical stability and logging bugs that had made earlier sessions unusable, restoring clean behavioural data capture, and shipped weekly iteration cycles alongside PhD researchers.

EDUCATION

BSc Computer Engineering · Universitat Pompeu Fabra (UPF), Barcelona

2022 – Jun 2026

Specialisation: Video Game Development & Interaction. Coursework: Game Engine Development (C++), Interactive Systems, Human-Computer Interaction, Narrative & Videogames, Data Structures & Algorithms.

SKILLS

Engines & Languages: Unreal Engine 5 (C++, ~6-month solo project) · Unity (3+ yrs) · C# · C++ · Python · Java · SQL · Git

Programming: Gameplay systems, AI (perception, state machines, navmesh), save/load, engine subsystems, shaders & post-process materials, tools, debugging & profiling

Also: Interaction & systems design · game balancing · GDD authoring · mixed reality & full-body interaction · HCI

Languages: English (C2) · Spanish (native) · Catalan (native)